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Transitional Epithelium is found lining the ureters, urinary bladder and part of the urethra. This epithelium allows stretching and is protective in function.

All cells of the epithelium are held together with little intercellular material. In most of the animal tissues, specialized junctions provide both structural and functional links between its individual cells. Three types of cell junctions are found in the epithelium and other tissues. These are called as tight, adhering and gap junctions. **Tight junctions** help to stop substances from leaking across a tissue. **Adhering junctions** perform cementing to keep neighbouring cells together. **Gap junctions** facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules.

3.3 Connective Tissue

Connective tissue develops from the **mesoderm** and is widely distributed in the body. There are four main classes of connective tissues. They are **connective tissue** (which includes fat and the fibrous tissue of ligaments), **cartilage, bones** and **blood.** Major functions of connective tissues are **binding** and **support**, **protection**, **insulation** and **transportation** of substances.

1. What type of connective tissue is damaged when one get cut on his index finger accidently?
2. The stored lipids are in the form of adipose tissue. Are they coloured? why?

Components of connective tissue

All connective tissues consist of three main components namely fibres, ground substance and cells. The **'Fibres'** of connective tissue provide support. Three types of fibres are found in the connective tissue matrix. They are **collagen**, **elastic** and **reticular** fibres. Connective tissue are of two types namely, **Loose connective tissues** (Areolar, Adipose and Reticular) and **Dense connective tissues** (dense regular, dense irregular and elastic)**. Specialized connective tissues** include cartilage, bone and blood.

Loose connective tissues

In this tissue the cells and fibres are loosely arranged in a semi fluid ground substances. For example the **Areolar connective tissue** beneath the skin acts as a support framework for epithelium and acts as a reservoir of water and salts for the surrounding body tissues, hence aptly called ***tissue fluid.*** It contains fibroblasts, macrophages, and mast cells (Figure 3.5).

Connective Tissues

Loose Connective Tissues	**Dense Connective Tissues**	**Specialised Connective Tissues**
1. Areolar Tissue	1. Dense Regular	1. Cartilage
2. Adipose Tissue	2. Dense Irregular	2. Bone
3. Reticular Tissue	3. Elastic	3. Blood